

Signals & Systems

BRANCH: ELECTRONICS (ECE)

COURSE CODE: EC201

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Signals & Systems (EC201) under the Electronics (ECE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Classification of Signals

Continuous-time vs Discrete-time signals, Energy vs Power, Periodic vs Aperiodic, Even vs Odd, Causal vs Non-causal. Basic operations: scaling, shifting, inversion.

Unit 2: LTI Systems

Properties of systems: Linearity, Time-invariance, Causality, Stability, Memory. Impulse response, Convolution integral, Convolution sum representation for LTI systems.

Unit 3: Fourier Analysis

Continuous-time Fourier Series (CTFS), Continuous-time Fourier Transform (CTFT). Frequency response of LTI systems. Discrete-time Fourier Transform (DTFT).

Unit 4: Laplace & Z-Transform

Laplace Transform: Region of Convergence (ROC), system transfer function, stability. Z-Transform: ROC, Inverse Z-transform, difference equations.

*Convolution: $y(t) = x(t) * h(t) = \int x(t-\tau)h(\tau)d_\tau$. Euler's Identity: $e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$.*

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers

